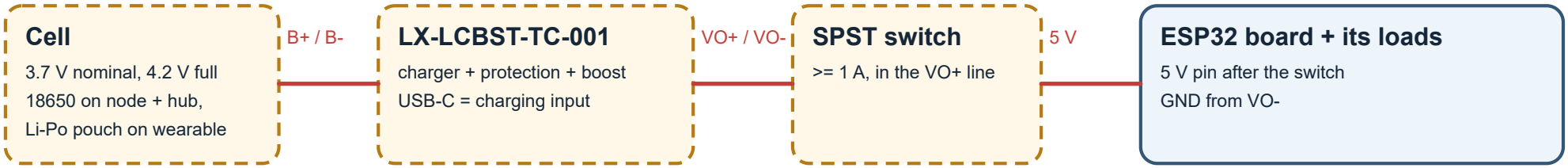
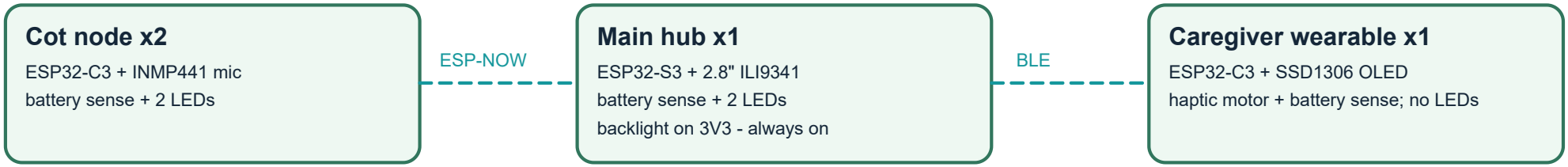
1 / CryNet system - power chain and links

Every unit has its own battery and its own ground. No wire ever joins two units.



One chain in all four units - only the loads differ:



Rules every unit follows:

1 - Switch OFF before charging. The cell still charges on B+/B-.

2 - Set the boost trimpot to 5.1 V with a meter before any board.

3 - Never power a board from USB and the boost at the same time.

Dashed teal = wireless link, no wire. Solid = a real conductor.

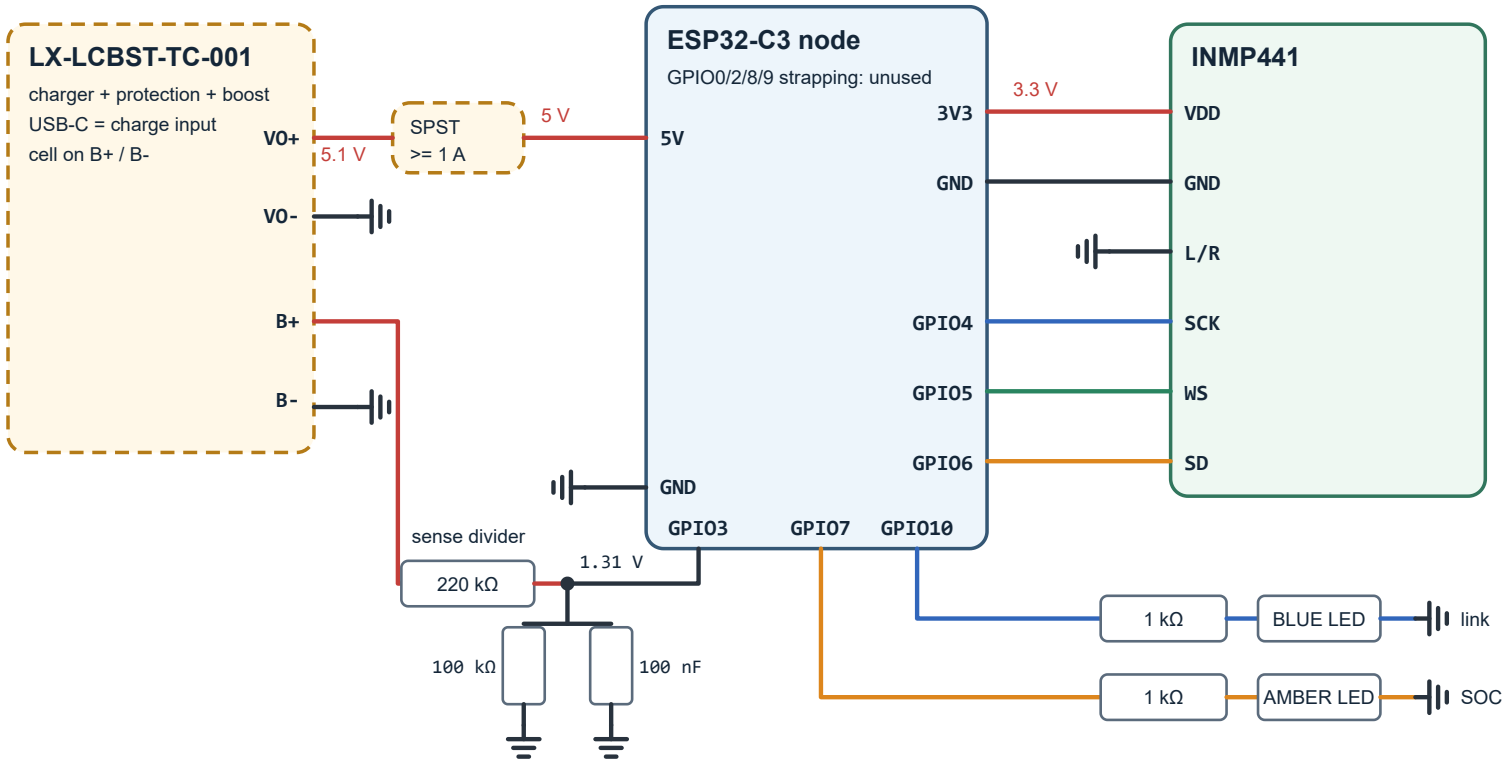
Proposed, bench-validation pending: the whole battery chain. Firmware status: on the node the battery divider and the GPIO10 blue link LED exist and run; the GPIO7 amber SOC LED does not (no GPIO7 in node_crynet.ino). The hub firmware has no ADC read and no LED output at all, so its divider and its two LEDs are plan only. The wearable has no LEDs - the OLED carries the state (it shows LINK or SCAN today).

Cells (D-81): an 18650 (3.7 V, 4.2 V full, 3000 mAh) on each node and the hub, because a node sits on a cot frame within a baby's reach and needs a cell that cannot be punctured; a 1000 mAh Li-Po pouch on the wearable, because an 18650 is 18 x 65 mm and 48 g and cannot go on a wrist. **Never LiFePO4** - the module charges to 4.2 V and an LFP cell tops out at 3.65 V. The module's ~1 A charger must come down to ~400-500 mA for the wearable's pouch (1 C is too high for it).

Runtime (D-82): the chain boosts to 5.1 V and the board drops that to 3.3 V through a *linear* regulator, which passes current rather than power - so the cell supplies roughly 1.8x the 3.3 V load current. That projects to about six hours for the wearable on a 1000 mAh pouch. These are projections, not measurements. Feeding the 3V3 pin from a 3.3 V buck-boost would roughly double it - a change to the power chain, deferred to the bench.

2 / Cot node - complete wiring (one per node, x2)

ESP32-C3 SuperMini + INMP441 + LX-LCBST + switch + battery-sense divider + 2 LEDs



Mic placement - the INMP441 is fixed-gain: mount it on the cot frame on a stiff lead, at least 30 cm from where the baby's head will be - never loose on the mattress.

Verified on the live system: mic wiring (3V3 / GND / L/R→GND, GPIO4 SCK, GPIO5 WS, GPIO6 SD) and the node firmware. The battery divider **is** implemented: node_crynet.ino uses PIN_BATTERY 3 (ADC1_CH3), a trust gate that only accepts 800-1500 mV at the pin, and scales the reading by 3.2 - so the 220 kΩ / 100 kΩ values above are the right ones. The GPIO10 blue link LED is implemented too (PIN_LINK_LED 10). GPIO1 stays spare.

Needs firmware on a branch: the GPIO7 amber SOC LED is a dead wire as it stands - there is no GPIO7 anywhere in node_crynet.ino, so nothing drives that LED yet. On USB power the bench node prints "battery none (USB, or the divider is not wired)" - that is expected, it is the divider trust gate refusing an out-of-range pin reading, not a fault.

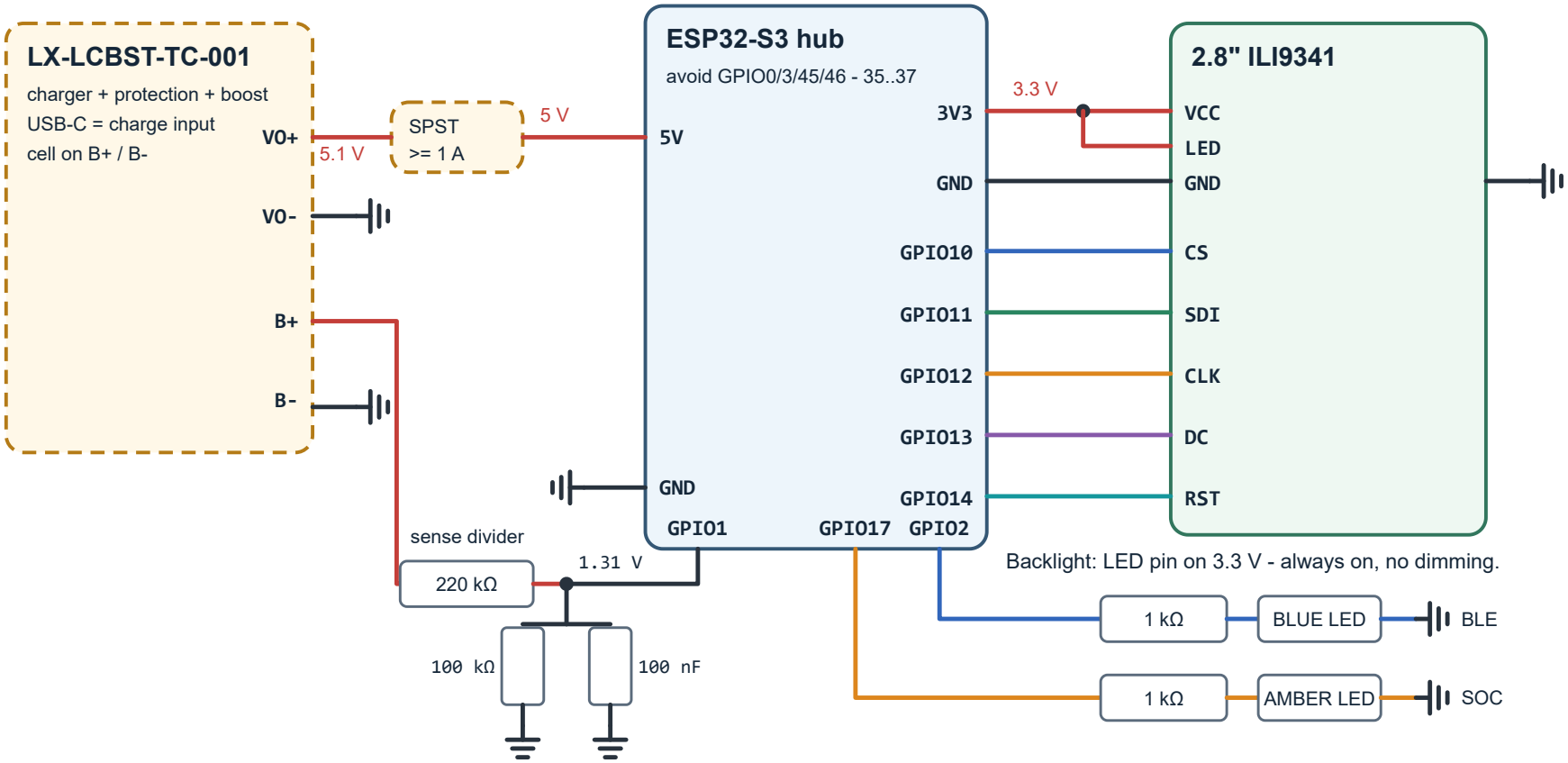
Cell (D-81): an 18650 Li-ion cell, 3.7 V nominal / 4.2 V full, 3000 mAh or better, **protected**, in a holder. A node sits on a cot frame within a baby's reach, so the steel can is the point - it cannot be punctured. **Never LiFePO4**: the module charges to 4.2 V and an LFP cell tops out at 3.65 V.

Test cry: the hub refuses a saturated window ("the window is CLIPPED"), so keep the phone at least 15 cm from the mic when you play a test cry through it.

Every ⏚ symbol is the same net: the module's VO- (battery negative through the module's protection FET). Wires join only where a dot is drawn. Pin names, not header positions. Repeat identically for node 2.

3 / Main hub - complete wiring (x1)

ESP32-S3-DevKitC-1 N16R8 + 2.8" ILI9341 (backlight on 3V3, always on) + LX-LCBST + switch + battery-sense divider + 2 LEDs



Verified on the live system: the TFT signal pins (CS 10, MOSI 11, SCK 12, DC 13, RST 14, VCC 3V3, GND). The panel's LED (backlight) pin is tied to the same 3.3 V net as the panel's VCC pin, so the backlight is permanently on: the glass cannot be dimmed and the display has no sleep mode. The hub firmware still drives GPIO4 as a backlight output (hub_display.c: PIN_BL 4, ili9341_backlight_set), so GPIO4 is left unconnected and the brightness control on the panel has no effect on the glass.

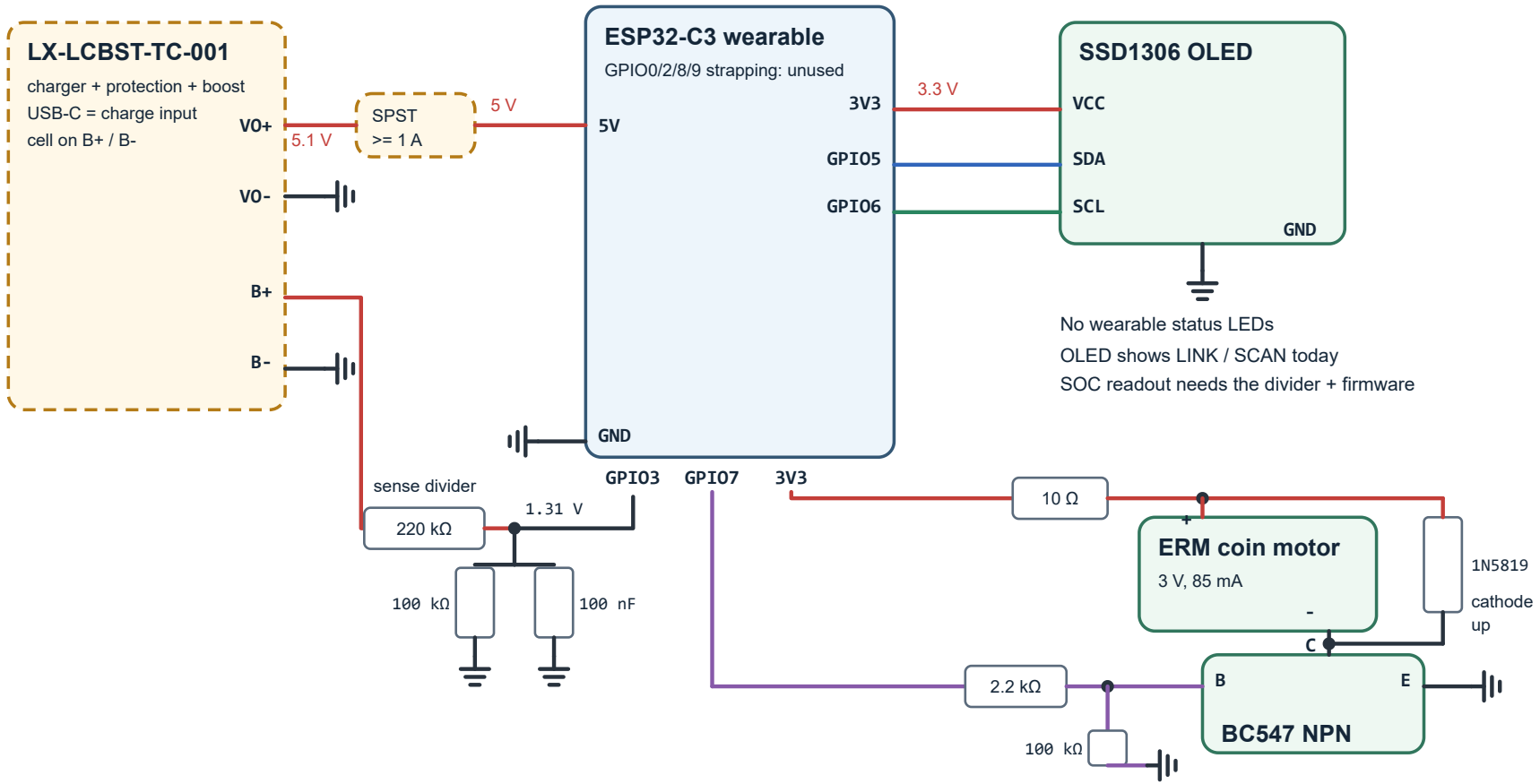
Plan only - the hub firmware does none of this yet: it has no analogReadMilliVolts at all and drives no LED. So the GPIO1 divider (ADC1_CH0), the GPIO2 blue BLE LED and the GPIO17 amber SOC LED are wiring plans, not working circuits. (BAT_CURVE / battery_band_advance in hub_display.c describe the NODES' reported cell voltage, not the hub's own cell.) The hub's peak current with TFT + radios is unmeasured - confirm the LX-LCBST holds 5.1 V during bursts before trusting it here.

Cell (D-81): an 18650 Li-ion cell, 3.7 V nominal / 4.2 V full, 3000 mAh or better, **protected**, in a holder - the hub draws the most current of any unit, so it is the cell most worth sizing carefully. **Never LiFePO4**: the module charges to 4.2 V and an LFP cell tops out at 3.65 V. The **divider is a fixed 220 kΩ / 100 kΩ pair** (x3.2) - no potentiometer in the battery path - a two-terminal pot cannot divide; a three-terminal one needs a per-unit ratio and drifts. The 100 nF is required.

GPIO35-37 are the Octal PSRAM bus - never wire them. The TFT signal wiring is the recorded working one for the existing module. The backlight is on 3.3 V and GPIO4 is not used. Every ⏚ symbol is the same net: the module's VO-.

4 / Caregiver wearable - complete wiring (x1)

ESP32-C3 SuperMini + SSD1306 OLED + LX-LCBST + switch + battery-sense divider + BC547 haptic drive (status on OLED)



Divider confirmed: 220 kΩ from B+ and 100 kΩ to ground give 4.2 V × 100/320 = 1.31 V at the ADC (the 100 nF filters it), 13 uA of divider current; the node firmware already uses this ratio (reading × 3.2). **The divider is fixed on purpose (D-81)** - no potentiometer in the battery path - a two-terminal pot cannot divide; a three-terminal one needs a per-unit ratio and drifts. The 100 nF is required. Motor: the 10 Ω caps the stall current at 3.3 V ÷ (R + 10 Ω) - about 69 mA for a 37 Ω coin ERM, inside the BC547's 100 mA; a lower-impedance motor needs 22 Ω or a 2N2222A. GPIO7 drives the switch in 250 ms pulses. **No LEDs are fitted.** The OLED carries the state and the firmware already shows LINK or SCAN on it. The battery/SOC readout needs the GPIO3 divider wired AND firmware that does not exist yet - branch work.

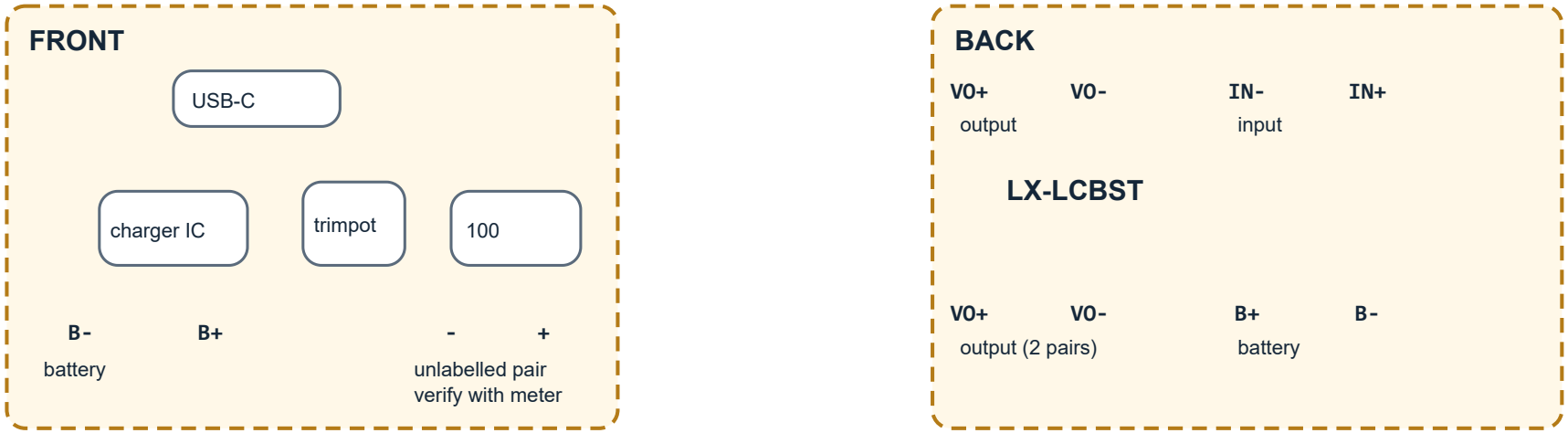
Cell (D-81): a 1000 mAh Li-Po *pouch* here, not an 18650 - an 18650 is 18 x 65 mm and 48 g and cannot go on a wrist, while the node and the hub take an 18650 because a cot frame can puncture a pouch. Same chemistry: 3.7 V nominal, 4.2 V full. **Never LiFePO4.** The module's ~1 A charger must come down to ~400-500 mA for this cell.

Runtime (D-82): 5.1 V boost, then the board's linear regulator down to 3.3 V - which passes current, not power, so the cell supplies roughly 1.8x the 3.3 V load current. That projects to about six hours on a 1000 mAh pouch; a projection, not a measurement. Feeding the 3V3 pin from a 3.3 V buck-boost would roughly double it.

Every ⏚ symbol is the same net; wires join only where a dot is drawn. The BC547 emitter is the motor's ground, and the flyback sits across the motor with the cathode up.

5 / LX-LCBST-TC-001 - pad map and bench procedure

From the user's front-and-back photograph - Chinese silk text translated



Bench procedure, before any ESP32 is connected:

1 - Cell on B+/B-: turn the trimpot until VO+/VO- reads 5.1 V on a meter, no load.

2 - USB-C in: measure the charge current with a series meter - the cell must allow it.

3 - Confirm the charge ends at 4.2 V.

4 - Switch off: confirm the load is dead and measure the battery drain (boost quiescent unknown).

5 - Run one node and compare it with the USB baseline.

No T pad on this board - the listing's T-shorting notes belong to other variants. Never short anything.

Unknown until measured: the actual charge current (the program resistor is unreadable in the photo), the boost quiescent drain when off, the protection thresholds, and what the front - / + pair connects to. Buy ONE module for validation; only then buy the remaining three (plus the wearable's, pending the same check).